



MINING DEALER BEST PRACTICE

Improvement in the Refrigerant Dryer Replacement Process of the Air Conditioner System for 793F Fleet

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

The dryer filter of the air conditioning system is a component composed of desiccant particles whose purpose is to retain solid impurities and absorb moisture. To identify when the filter needs to be changed, just examine the inlet and outlet pipes of the filter, the temperature must be the same in both tubes, these checks are carried out in the preventive maintenance of the 793F. An analysis of the downtime drivers of the PM process identified that a large amount of time is spent with refrigerant dryer replacement, due to the complexity of the component connections.

2.0 Best Practice Description

This BP provides information for installing a quick connect coupling for the refrigerant dryer (257-3226) of the air conditioning system of a 793F. Without a quick connect coupling, the refrigerant gas is drained (exhausted) by not having a check valve. Thus, it is always necessary to carry out the refrigerant recovery fully, which increases the maintenance time and costs.

3.0 Implementation Steps

The implementation of this BP consists of replacing the rigid piping interconnecting the filter dryer with a quick connect coupling hose system.

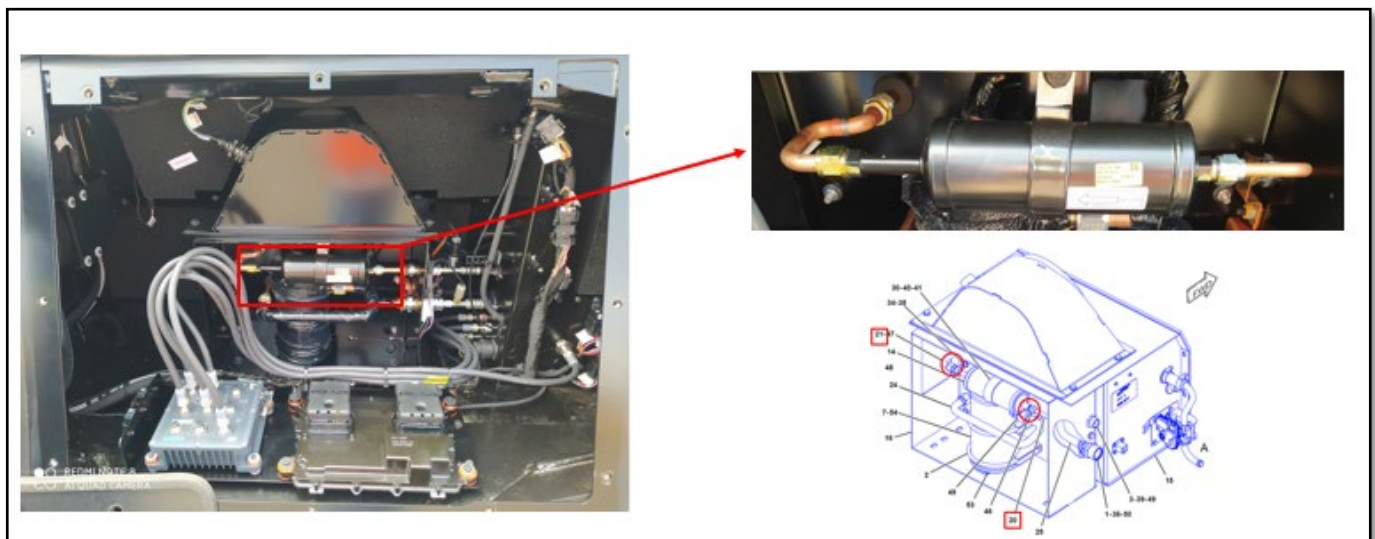


Figure 01 – Refrigerant Dryer as Shipped on 793F.

- Collect the refrigerant gas using the PN 285-7352 tool, remove the PN 357-2994 tube and the 257-3226 filter dryer.

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- Assemble the hose assembly according to the specifications:
 - Hose 541-7161 – 600 mm
 - 478-5446 - 01 Terminal
 - 320-1720 – 01 Terminal
- Order the following parts:
 - 02 quick coupling PN 103-6100
 - 01 Filter Dryer PN 3E-3535



Figure 02 - Applied improvement using quick coupling system.

3.1 Steps Summary:

1. Layout definition (Line Routing) for installation of air conditioning system lines;
2. Acquisition of parts needed for installation;
3. Installation of new dryer and air conditioner lines to charge the refrigerant on the system.

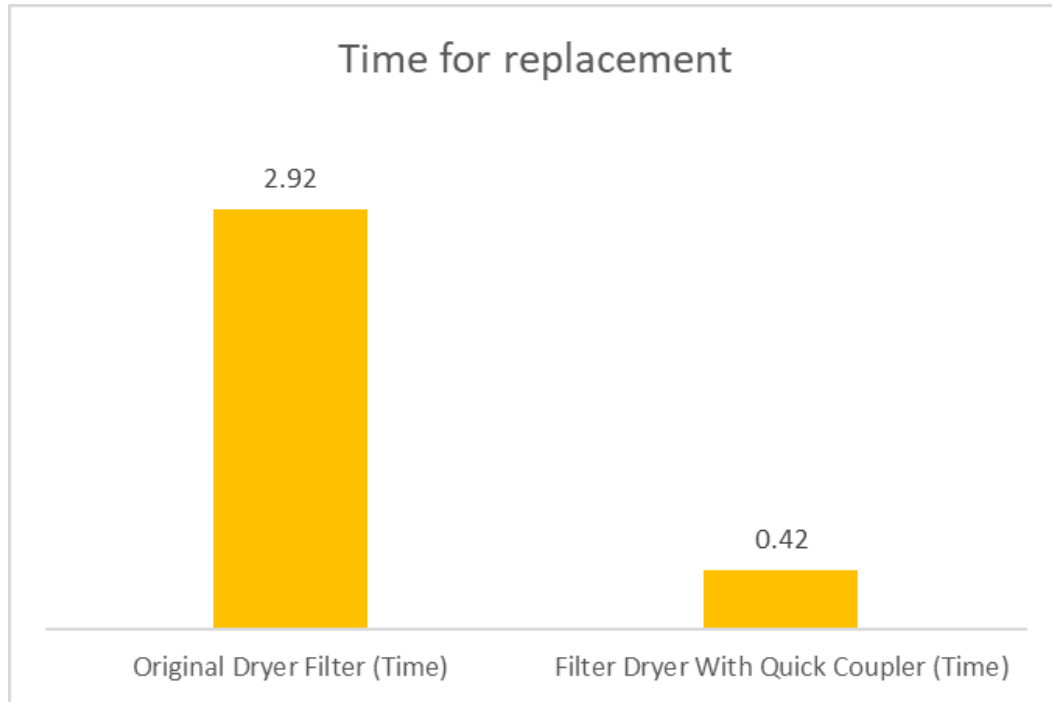
4.0 Benefits

- Maintenance cost reduction
- Contamination control
- Maintenance time reduction

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4.1 Save with implementation:

Original Dryer Filter (Time)	Filter Dryer With Quick Coupler (Time)
2.92	0.42



5.0 Resources Required

- Hose 5417161 – 600 mm.
- 4785446 - 01 Terminal.
- 3201720 – 01 Terminal.
- 02 quick coupling PN 103-6100.
- 01 Filter Dryer PN 3E-3535.

6.0 Supporting Attachments / References

In-Line Refrigerant Dryer - Remove and Install (SENR5664-35).

Receiver Dryer (Refrigerant) - Replace (SEBU8751-22).

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7.0 Related Best Practices

None

8.0 Acknowledgements

This Best Practice was written by:

Wender Madeira

wender.madeira@sotreq.com.br

Phone: 55-31-98885-7675

Belimar Costa

belimar.costa@sotreq.com.br

Phone: 55-31-99967-0731

Special thanks in development and implementation to:

Claudinei Ataide

claudinei.ataide@sotreq.com.br

+55-31-97188-7413

Paulo Vieira

Paulo.vieira@sotreq.com.br

+55-31-97166-3280

Anderson Silva

Anderson.silva@sotreq.com.br

55-31-996957094

Thiago Assis

Thiago.assis@sotreq.com.br

+55-31-98887-4301

Gustavo Lage

gustavo.lage@sotreq.com.br

+55-31-98675-8296

Claudio Cunha

Claudio.cunha@sotreq.com.br

Phone: 55-31-99929-3361

Best Practice Champion:

Fellipe Viana

fellipe.viana@sotreq.com.br

Phone: +55 31 99194-3974

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